Algorithms 2

Vertex-connectivity in graphs

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1. Connectivity

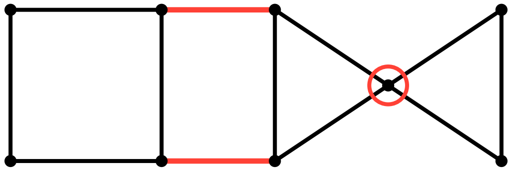
Definition 1.1.1

- For a graph G, write

$$C(G) := \text{number of connected components of } G$$

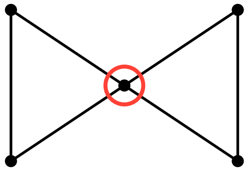
Definition 1.1.2

- Let $X \subseteq V(G) \cap E(G)$.
- If $C(G - X) > C(G)$, then X is called a *disconnector*



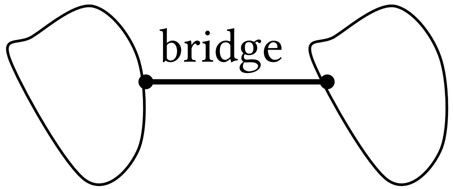
1.1 Graph disconnectors

if $X \subseteq V(G)$, then X is a *vx-disconnector*.



- vx-disconnectors of size 1 are called *cut-vxs*.

if $X \subseteq E(G)$, then X is a *edge-disconnector*.



- edge-disconnectors of size 1 are called *bridges*.

- The ends of bridges are cut-vxs.
- The opposite is **not** true!

Observation 1.1.3 (H.W.)

$$e \in E(G) \text{ is a brige} \Leftrightarrow e \text{ does not lie on a cycle of } G$$

2. Whitney's Theorem

2.1 Whitney's Theorem

Theorem 2.1.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } \; G \; \text{ lies on a common cycle.}$$

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proof $\Leftarrow$.

- Assume G has a cut vertex, denoted by v.

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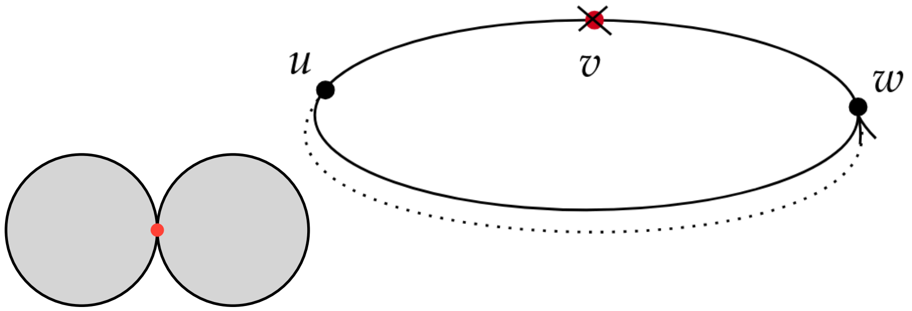
proof $\Leftarrow$.

- Assume G has a cut vertex, denoted by v.
- v seperates the graph into 2 components C_1, C_2.
- Take $w \in C_1, u \in C_2$ both should hold true



1. A cycle $C := wx_1 ... x_\ell uw$ exists in G.
2. If we remove v then u and v are disconenected.

contridition: If $v \in C$ then removing v or any vertex in C wont seperate w from u, and if $v \notin C$ then the path $wx_1 ... x_\ell u$ still exists.

2.1 Whitney's Theorem

Theorem 2.1.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } \; G \; \text{ lies on a common cycle.}$$

$\Rightarrow$

- Suppose G has no cut-vxs.
- We prove by induction on

$$\delta_{G(u,v)} := \text{length of the shortest path u} \rightsquigarrow \text{v in } G$$

- Basis:
  - Let $u, v \in V(G)$ have $\delta_G(u,v) = 1$
  - If $uv \in E(G)$ is a brigde, the u, v are cut-vxs. **contridiction**.
  - From the observation uv lies on a cycle.

Observation 2.1.3 (H.W.)

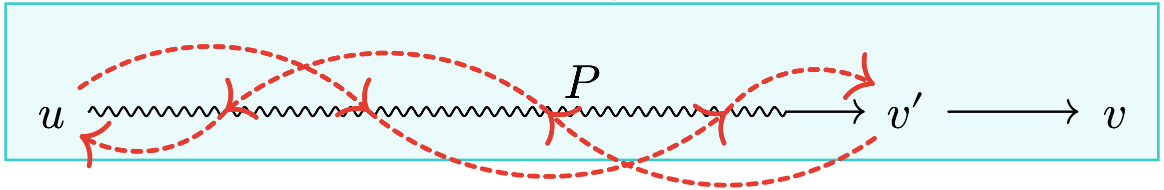
$$e \in E(G) \text{ is a brige} \Leftrightarrow e \text{ does not lie on a cycle of } \; G$$

Theorem 2.1.4 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

- Suppose G has no cut-vxs.
- Suppose that the claim hold true for all pairs of verecies u, v such that $\delta_{G(u,v)} < k$
- Consider a pair $u, v \in V(G)$ satisfying $\delta_G(u, v) = k$
- Let P be the shortest path $u \rightsquigarrow v$, and let v' denote the vertex before v on P.
- As $\delta_G(u, v') = k - 1$, there is a cycle C containting u, v'



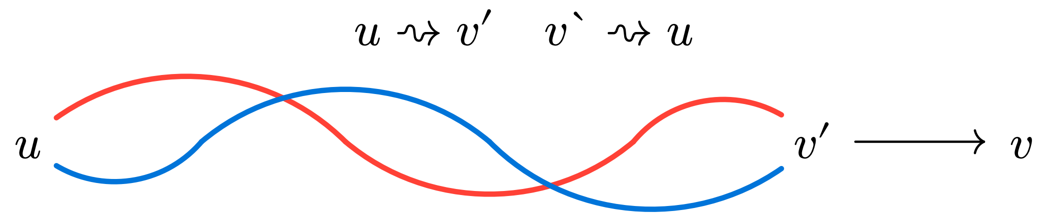
- If $v \in V(C)$ we are done

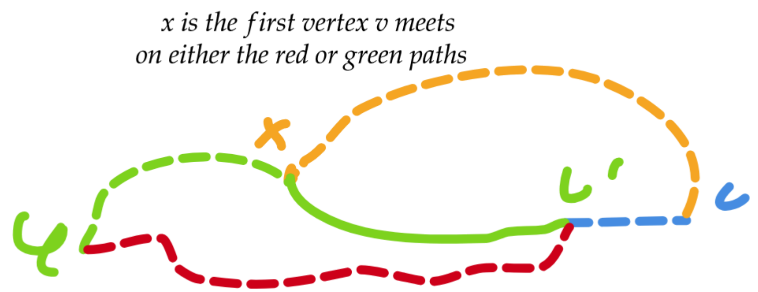
Theorem 2.1.5 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

- Otherwise, we can divide C into 2 arcs

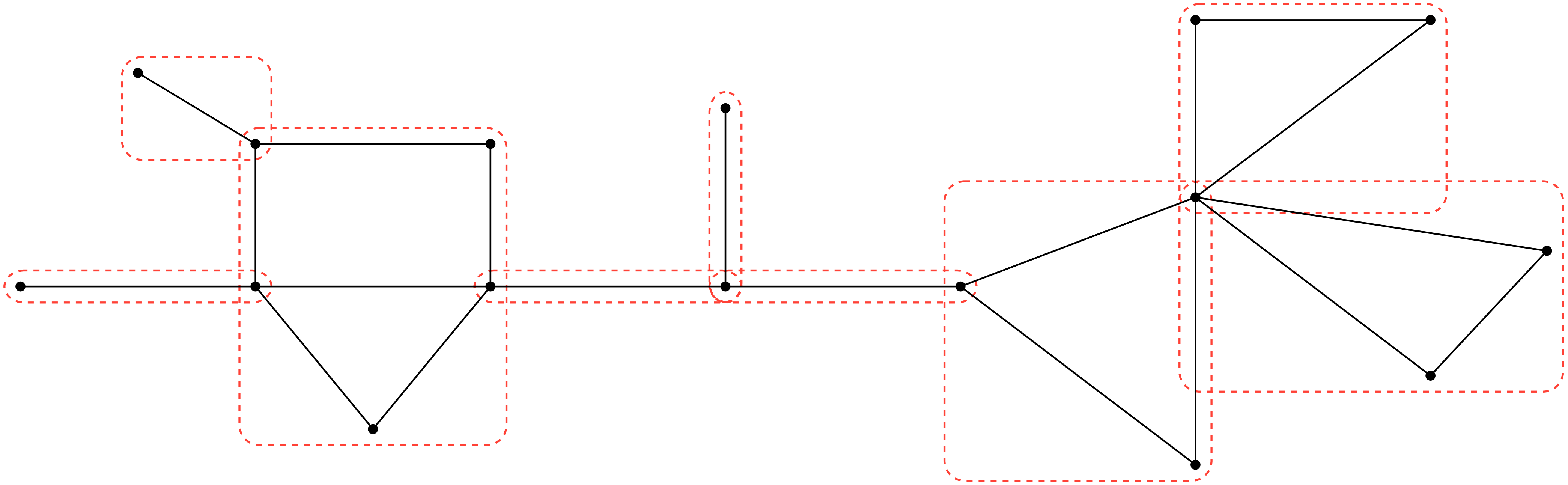




- As G has no cut-vxs, $G - v'$ is still connected,
- There is a path from v to a vertex x that lies on one of the arcs)
- W.L.O.G, let $x \in u \rightsquigarrow v'$ **take the shortest path from v to any of the arcs!**.
- Then the cycle $u \rightsquigarrow x \rightsquigarrow v \to v' \rightsquigarrow u$ is a cycle, and contains both u and v.

3. Consequences of Whitney's Theorem

Definition 3.1.1 A maximal connected subgraph of G containing no cut-vxs is called a block of G



3.2 Consequences of Whitney’s Theorem

Theorem 3.2.1 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } \; G \; \text{ lies on a common cycle.}$$

- Any two blocks of G meet in at most one vertex.
- If two block do meet, their intersection vertex is a cut-vertex.
- The blocks of G partition $E(G)$ (H.W.).
- Every cycle of G lies in precisly one block of G(H.W.).

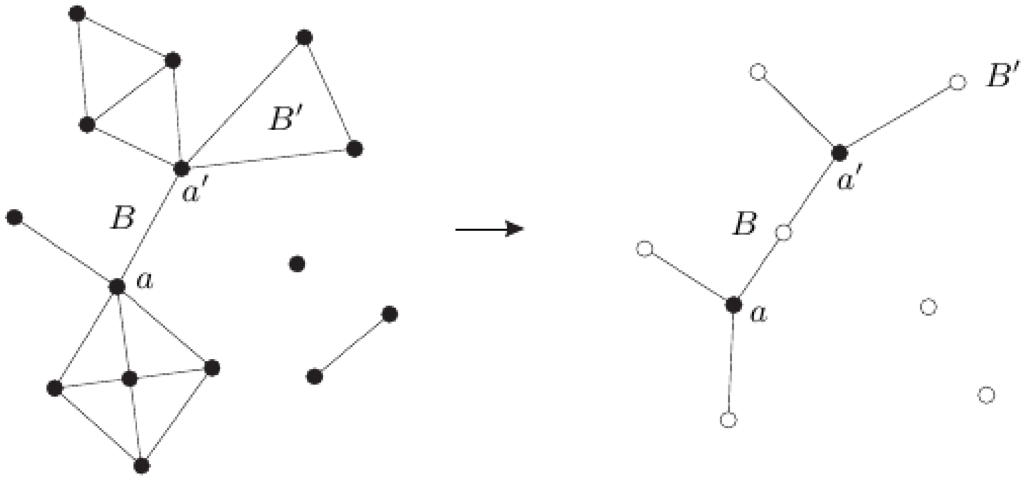
Given a graph G define the following auxiliary graph:

- $B(G) := \{B : B \subseteq G \text{ is a block of G}\}$
- $C(G) := \{v : v \in V(G) \text{ is a cut-vx of G}\}$

Let $\mathrm{BC}(G)$ denote the follwing graph

- The vertices of $\mathrm{BC}(G)$ are $B(G) \cap C(G)$
- The edges of $\mathrm{BC}(G)$ are
$$\{\{B, v\} : B \in B(G), v \in C(G) \text{ and } v \in B\}$$

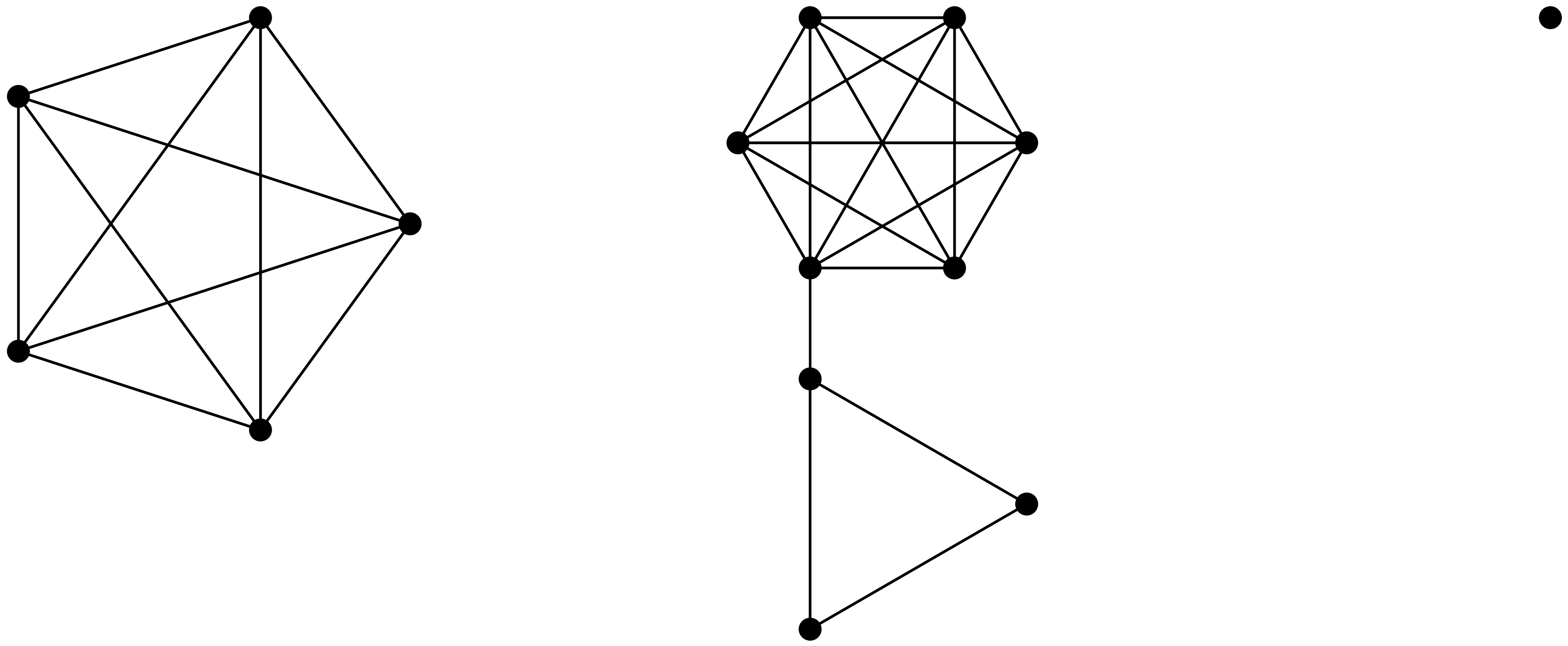
Theorem 3.3.1 if G is connected then $\mathrm{BC}(G)$ is a tree



4. Assesing connectivity

4.1 Assesing connectivity

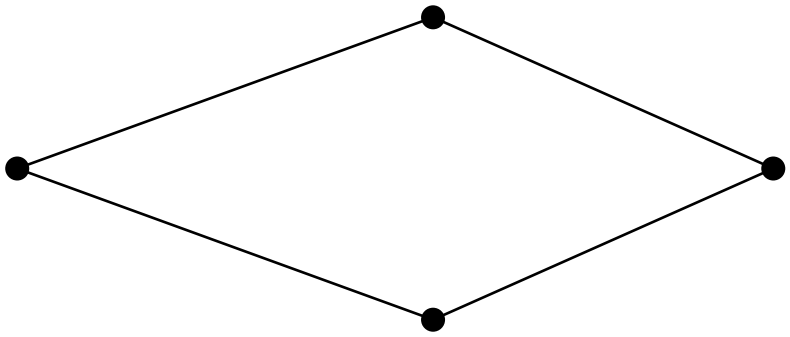
Question 4.1.1 Given these graphs, which one is more connected?



4.2 Definition I

Definition 4.2.1

- $0 \leq k \in \mathbb{N}$
- G - graph with $v(G) > k$
- If $G - X$ is conncted $\forall X \subseteq V(G), |X| \leq k-1$ then G is called *k-connected*



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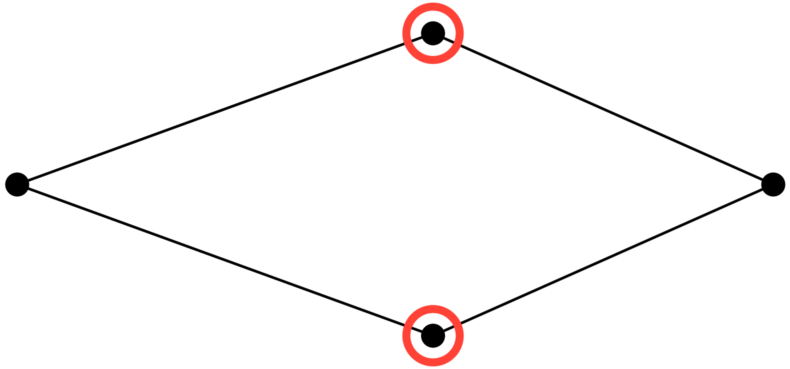

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- To be k-connected, $v(G) > k$ must hold

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- But our definition K_1 is not connected!

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- What about K_1?
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- But our definition K_1 is not connected!
- We'll fix this later

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- $\kappa(G) \geq 1$

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- $\kappa(K_r) = r - 1$

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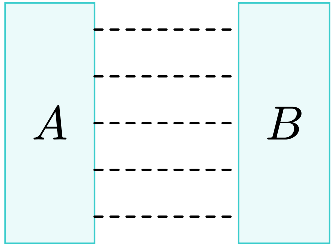
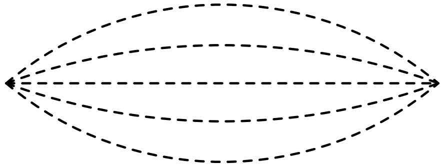
- $\kappa(K_r) = r - 1$

Again $\kappa(K_1) = 0$, is annoying, can we fix it?

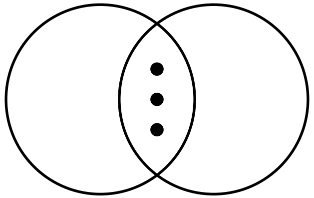
Definition 4.4.1

- G a graph
- $A, B \subseteq V(G)$
- let $\rho_G(A, B) :=$ the number of vx-disjoint (A, B)-paths





Here $|A| = |B| = 1$



$A \cap B \neq \emptyset$ also can happen

Remark

Vertices in A and B counts toward the vertices of the paths.
If $|A| = 1$ or $|B| = 1$ then we count the number of paths from the neibours of A or B respectively.

4.5 Definition II: Path connectivity

Definition 4.5.1 (Path connectivity) G is *k-connected* if $\rho_G(u,v) \geq k \quad \forall \{u,v\} \in \binom{V(G)}{2}$

Definition 4.5.2 (Vertex connectivity) Graph G with $v(G) > k > 1$ is *k-connected* if for every $X \subseteq V(G)$ with $|X| \leq k-1$ the graph $G - X$ is connected. If $v(G) = 1$ then G is *1-connected*.

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So whats better: vertex connectivity or path connectivity ?

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So whats better: vertex connectivity or path connectivity ?

Menger's theorem tells us that they are the same!

4.6 Our definition

Definition 4.6.1 The minimum size of a vx set X such that $G - X$ is disconnected or has a single vx is called the *vx-connectivity* of G and is denoted by $\kappa(G)$

Definition 4.6.2 A graph G is called *k-connected* if $\kappa(G) \geq k$

Remark

Other than the complete graph:

$$\text{A graph is } k\text{-connected} \Leftrightarrow \text{all of its vx-cuts are of size} \geq k$$

The complete graph has no vx-cuts of any size, this is where the 2nd part of the definition comes in

5. Menger’s theorem

5.1 Menger's theorem

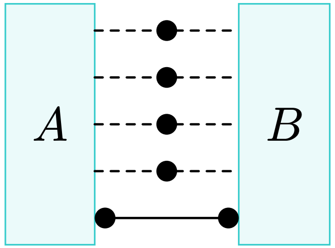
Definition 5.1.1
- $A, B \subseteq V(G)$
- Denote by $\kappa_G(A, B)$ the size of the **least** (A,B)-vx-disconnetor

Theorem 5.1.2 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\kappa_G(A, B) = \rho_G(A, B)$$

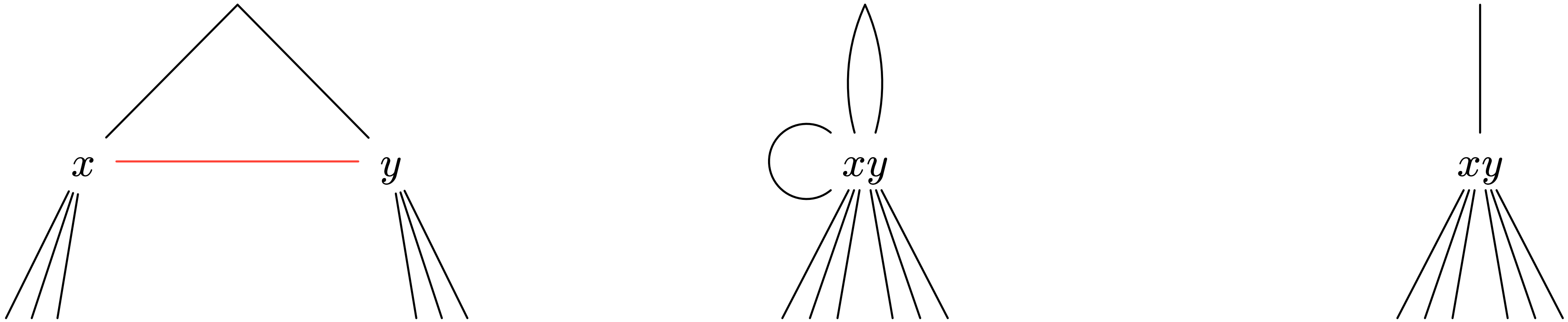
- You know this by the name "min-cut-max-flow"
- $\kappa_G(A, B) \geq \rho_G(A, B)$ is trivial
- We need to prove

$$\kappa_G(A, B) \leq \rho_G(A, B)$$



5.2 Edge contractions

- Consider the edge $e = xy$
- To constract e:
  - remove e
  - merge x and y into one vertex v_{xy}
  - connect v_{xy} to $N_G(x) \cap N_G(y)$, remove any dupplicant edges that may arise



- Denote by G/e the graph resualting from contracting e

5.3 Menger's theorem

Theorem 5.3.1 (Menger's theorem) Let G be a graph and let $A, B \subseteq V(G)$ non empty. Then

$$\underbrace{\kappa_G(A,B) = \rho_G(A,B)}_{\text{The goal is to show } \leq}$$

- Set $\kappa := \kappa_G(A,B)$ and $\rho := \rho_G(A,B)$
- Proof by induction on $e(G)$
- If $e(G) = 0$ the graph is disconnected
  - $\rho = |A \cap B| = \kappa$

Goal : Show that $\kappa_G(A, B) \leq \rho_G(A, B)$.

I.H: For every graph G' with $e(G') < e(G)$ and every $A', B' \subseteq V(G')$ non empty, we have $\kappa_{G'}(A', B') \leq \rho_{G'}(A', B')$

- Assume $e(G) > 0$, let $e = xy \in E(G)$ be an arbitrary edge
- G/e satisfies the I.H. but A and B are not defined consistently in $G \setminus e$. **(What if x or y is in A or B?)**.

  Define $A_e \subseteq V(G/e)$ as follows
  - If $x, y \notin A$, then $A_e := A$
  - if $x \in A$,$y \notin A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
  - if $x \notin A$,$y \in A$, then $A_e := (A \setminus \{x\}) \cup \{V_e\}$
  - if $x, y \in A$, then $A_e := (A \setminus \{x, y\}) \cup \{V_e\}$
- Define B_e in the same manner.

Case 1: Suppose that G/e has κ vx-disjoint (A_e, B_e)-paths

- **Goal:** Show that every such path exists in G.
  - Any path not containing v_e exists in G.
  - suppose that there is a path in the linkage containing v_e.

xy can be expanded in the path

$$\text{- - - - - - - -} \in N_G(x) \text{———} x \text{———} y \text{———} \in N_G(y) \text{- - - - - - - -}$$
$$\text{- - - - - - - -} \in N_G(y) \text{———} y \text{———} x \text{———} \in N_G(x) \text{- - - - - - - -}$$

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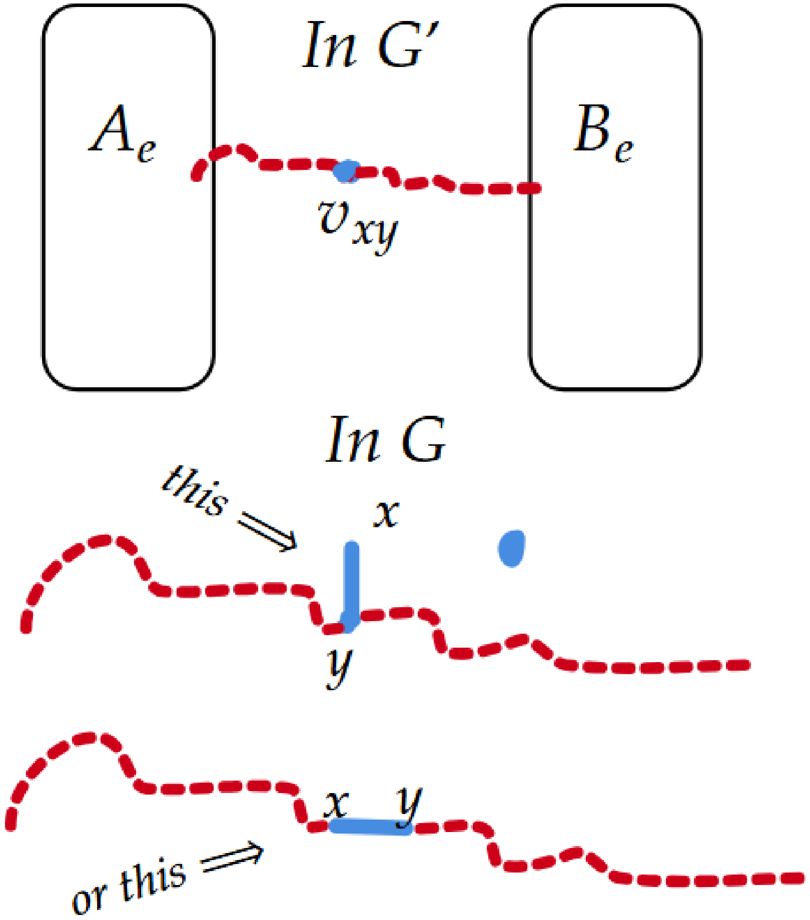
- In the second case, only one of the vertices lie on the original path.

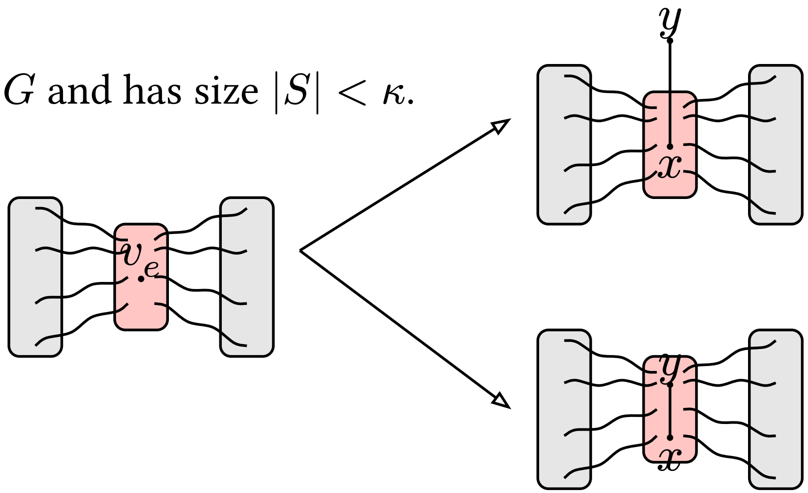


Claim 5.4.1 If S is vx-disconnector of (A_e, B_e) in G/e then a vx-disconnector of (A, B) in G exists where

$$S_G \subseteq (S \setminus v_e) \cup \{x, y\}.$$

Case 2: $\rho_{G \setminus e}(A_e, B_e) < \kappa$

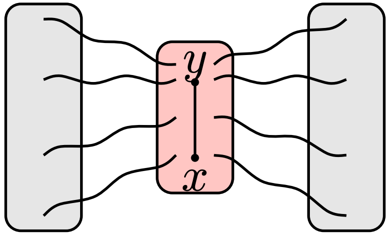
- By the I.H $\kappa_{G \setminus e}(A_e, B_e) < \kappa$.
- Let S be a vx-disconnector of (A_e, B_e).
- $v_e \in S$ must hold:
  - If $v_e \notin S$, then S is also a vx-disconnector of (A, B) in G and has size $|S| < \kappa$.
- Any (A_e, B_e)-vx-disconnector in G/e is of size $\kappa - 1$.
  - Otherwise $|S_G| \leq |S| + 1 < \kappa - 1 + 1 = \kappa$.



5.4 Menger's theorem

We are now in this position:

- The edge xy lies inside the minimum vx-disconnector of (A, B) in G.



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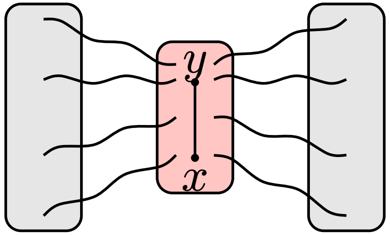
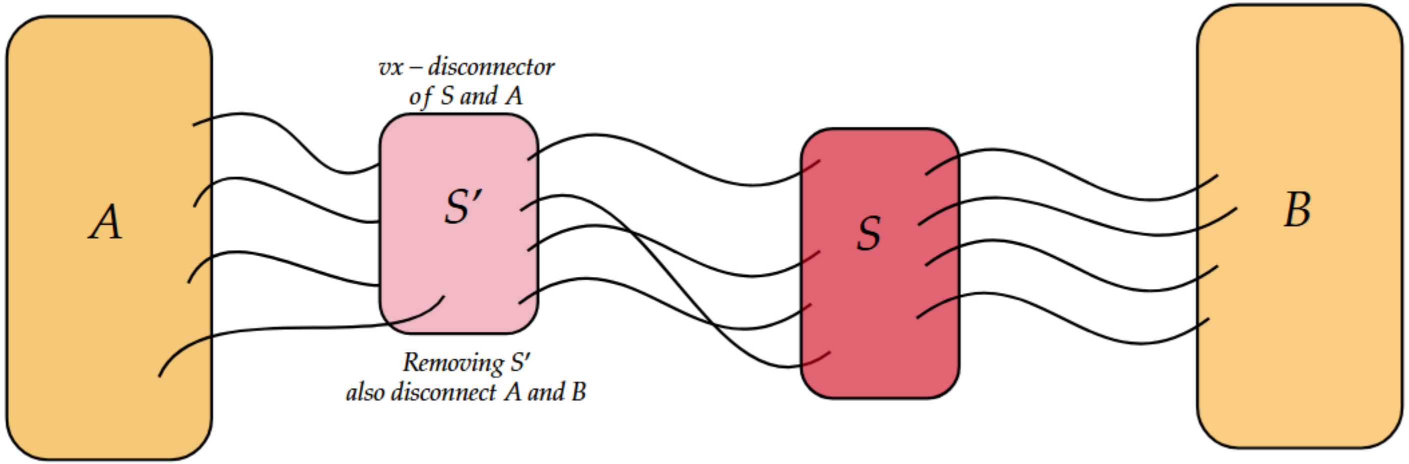
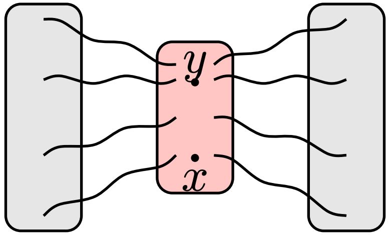


Observation: The size of the least vx-disconnector of (A, S) and (S, B) is at least κ.

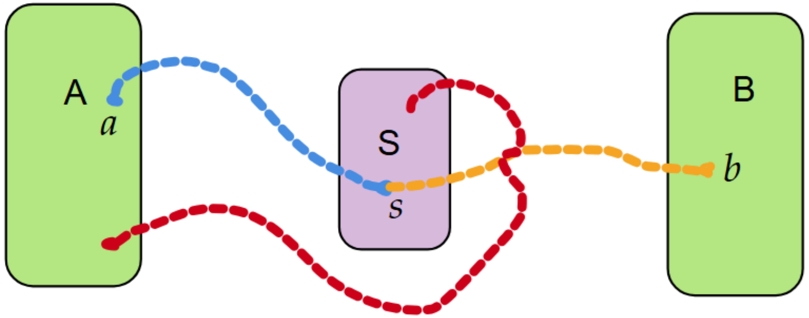


- Look at $G - e$: **(removing the edge e rather than contracting it!)**
- By the induction hypothesis.

$$\rho_{G-e}(A, S_G) = \kappa_{G-e}(A, S_G) \geq \kappa \quad \text{and} \quad \rho_{G-e}(A, S_G) = \kappa_{G-e}(S_G, B) \geq \kappa$$



- There must exists κ vx-disjoint (A, S_G)-paths and κ vx-disjoint (S_G, B)-paths in $G - e$.
- For any path $a \rightsquigarrow s$ and $s \rightsquigarrow b$, where $a \in A$, $s \in S$, $b \in B$.
  - The path $a \rightsquigarrow s \rightsquigarrow b$ is vx-disjoint (A, B)-path in G.



Red path cannot occur in G,
As then S doesnt disconnect A and B.

6. Implication of Menger’s theorem

6.1 Implication of Menger's theorem

Corollary 6.1.0.1 (H.W.)

- Let G be a graph. Then

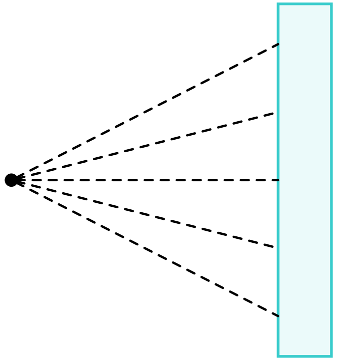
$$\kappa(G) = \min\{\rho_G(u,v) : u, v \in V(G), u \neq v\}$$

Lemma 6.1.1 (The Extension Lemma H.W.)

- G - k-connected graph
- H obtained from G by:
  - Adding a new vertex
  - Connecting the new vertex to k arbitrary vertecies
- Then H is k-connected,

6.1 Implication of Menger's theorem

- Given $x \in V(G)$ and $B \subseteq V(G)$ s.t. $x \notin B$, we write *(x, B)-fan* to denote:



Lemma 6.1.2 (Fan Lemma H.W)

- G - k-connected graph
- $x \in V(G)$ and $B \subseteq V(G) \setminus \{x\}$

Then G has an (x, B)-fan if size $\min\{k, |B|\}$

Theorem 6.2.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k-connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constaining S

- By induction on k.
- for $k = 2$, the claim holds by whitney's theorem.

Theorem 6.2.2 (Whitney's Theorem)

- G - a connected graph with $v(G) \geq 3$

$$G \text{ has no cut-vxs} \Leftrightarrow \text{any 2 verticies of } G \text{ lies on a common cycle.}$$

6.2 Dirac's theorem

I.H: If G is a $k-1$ connected graph then for every $S \subseteq V(G)$ with $|S| \leq k-1$ there is a cycle containing S.

- Assume $k \geq 3$
  - Let $x \in S$ be arbitrary and set $T := S \setminus \{x\}$.
- |T| <= k-1.
- G is k-connected so $G - x$ is at least $k-1$ connected.
  - By the I.H. there is a cycle C containing T.
- let F be an (x, C)-fan in G of size $\min\{k, v(C)\}$.
- T partitions C into $k-1$ arcs.
- At least two of the paths in the fan land in the same arc on C.

Theorem 6.2.1 (Dirac's Theorem)

- $2 \leq k \in \mathbb{N}$
- G - k-connected graph
- $S \subseteq V(G)$ such that $2 \leq |S| \leq k$

Then G contains a cycle constaining S